Predicting Zero-Shot Video-Tracker Transfer: Methods, Results, and Engineering Log

Progress report — "will-it-track"

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Abstract

We are testing whether the zero-shot tracking performance of a frozen promptable video model (SAM 3) on an *unseen* species or place can be predicted *before running it*, from four label-free "distances" (taxonomic, visual, environment, temporal), and whether the two halves of tracking — *finding* the animal (pDetA) and *following* it (pAssA) — are governed by different factors. This document records, in plain language, exactly what has been built and run so far, what the results say, and every practical and statistical problem we hit together with the fix we applied. The headline so far is an *honest null*: on the two splits tested, the distances do not predict SAM 3's per-cell detection/association to a degree any valid test supports. This is a legitimate scientific result, and it points clearly at the next experiment (a powered species hold-out), which is running now.

1 Aim and hypotheses

Conservation teams point promptable trackers at new species and new camera-trap sites constantly, but nobody can say in advance whether the model is trustworthy for a given animal in a given place. We try to fill that gap with a *reliability estimator*: given only label-free distances, predict pDetA and pAssA with honest confidence intervals. The guiding hypotheses are:

H1 (detection). pDetA falls with **species novelty** (taxonomic + visual distance).

H2 (association). pAssA falls with **environment difficulty** (scene distance, day/night/IR, clutter), largely regardless of species.

H0 (null). Distance explains little variance — *still a valid result*, and one that would redirect the project toward representational probing.

The goal is explicitly **not** to make the tracker score higher; it is to *predict* and *explain* its transfer.

2 Data

We use **SA-FARI** (Meta × Conservation X Labs, 2025), the largest open multi-animal tracking dataset: 11,609 camera-trap videos, 99 species with a full 7-level taxonomy, 741 locations on 4 continents, and per-video timestamps. Annotations are on 6 fps frames (mean ≈85 frames per clip). Crucially, SAM 3 is **frozen and zero-shot** on all of SA-FARI, so the train/test split is only our own reference anchor for measuring distances — not anything the model learned.

3 Method

The pipeline has five stages; each is a small, config-driven module.

3.1 Inference

A frozen SAM 3 promptable tracker is run over every probe of a split. For each video we give a text prompt (the species name, e.g. `"impala"`) and the model segments and tracks every matching animal across the clip. Predictions are written one JSON per video, so the run is *resumable*. Hard negatives (a query for an absent species) are kept — they should return nothing.

3.2 Scoring

We score with the *official* SA-Co / VEval evaluator (never re-implemented). The promptable HOTA score decomposes cleanly into the two parts we analyse separately:

$$\mathsf{pHOTA} = \sqrt{\mathsf{pDetA} \times \mathsf{pAssA}}, \qquad \mathsf{pDetA} = \text{did it find the animals?}, \quad \mathsf{pAssA} = \text{did it keep each identity over}$$

Scores are recorded per **cell** — a (species, location, time) group — together with the *support* behind each score (number of frames, masklets, videos).

3.3 The four label-free distances

Every distance is computable *without any label of the target species*, which is what makes a before-running prediction possible.

Distance	What it measures	Granularity
Taxonomic	Tree distance (Kingdom→Species) to the nearest reference species	per species
Visual	$1 - \cos$ between DINOv2 embeddings of the *mask-cropped animal* and the nearest reference species' prototype	per species
Environment	$1 - \cos$ between DINOv2 embeddings of the *masked-background scene* and the nearest reference location's prototype; plus a night/IR flag and a clutter score	per location
Temporal	Gap (in years) between the probe footage and the reference footage	per cell

Table 1: The four distances. Visual and environment reuse a shared, cache-first embedding pipeline.

3.4 The two experiments (splits)

Split B — location hold-out (secondary, tests H2). Reference = official train split, probe = test split. Camera sites are unseen; species are largely shared, so the taxonomic distance is ≈ 0 and the environment/temporal distances vary. This is the "does the scene matter?" experiment.

Split A — species hold-out (primary, tests H1). Leave-*one*-species-out over the pooled present set: each probe species' distance is to the nearest *other* species, which makes taxonomic novelty a genuine *graded* axis. This is the "does novelty matter?" experiment.

3.5 Modelling

For each target (pDetA, pAssA) we fit a support-weighted logit-link GLM (a fractional-logit model on the bounded score), with the four distances plus the scene covariates and a log(support) covariate so that "rare" is never mistaken for "far". Predictors are standardised. We then:

- report **group-cluster-bootstrap** confidence intervals (refit while resampling whole species / locations) — see the confidence-interval issue below;

- run **grouped cross-validation** (leave-one-species-out and leave-one-location-out) and test the out-of-sample gain over a mean predictor with a **paired group-bootstrap** p-value;
- partition variance across factors (dominance analysis) and check collinearity (VIF).

4 Results so far

4.1 Gate 1 — measurement (is SAM 3 behaving sensibly?)

Zero-shot SAM 3 on the SA-FARI test split reproduces the published ballpark. The dataset-level (micro) mask pHOTA is ≈ 0.65 with a real detection–association gap. Table 2 reports the per-cell (macro) support-weighted means we feed to the models; these are lower than the micro number because 78% of cells are single-object clips where pAssA≈pDetA (finding and following coincide when there is only one animal).

Metric (mask, per-cell macro mean)	Value
pDetA	0.527
pAssA	0.546
pHOTA	0.533
Cells scored	346 of 1447

Table 2: Gate 1 measurement on the test split. The 1101 unscored cells are hard negatives (species absent, no ground-truth track), correctly excluded from the HOTA fit and reserved for a separate false-positive analysis.

4.2 Gate 2 — location split (Split B): a null

After correcting the inference (the confidence-interval issue below), **every distance coefficient's confidence interval spans zero**, and the out-of-sample gain over a mean predictor is **not significant** on either cross-validation scheme (Tables 3–5). The large-looking taxonomic coefficient (−0.36) is a fragile artifact of this split (see the novelty-axis issue below), not a graded effect.

Factor	pDetA coef. [95% CI]	pAssA coef. [95% CI]
Taxonomic distance	−0.362 [−3.87, 0.09]	−0.368 [−3.40, 0.08]
Visual distance	−0.144 [−0.50, 0.14]	−0.122 [−0.49, 0.18]
Environment distance	+0.042 [−0.19, 0.24]	+0.013 [−0.23, 0.21]
Clutter	−0.066 [−0.38, 0.29]	−0.087 [−0.41, 0.28]
Night / IR	−0.405 [−0.94, 0.11]	−0.462 [−1.02, 0.11]
log(support)	−0.216 [−0.51, 0.17]	−0.224 [−0.53, 0.18]

Table 3: Split B (location) standardised coefficients with group-cluster-bootstrap CIs. Every distance spans zero. The temporal gap is omitted: it is constant on this split and carries no information.

4.3 Gate 2 — species hold-out (Split A, preview): a cleaner null

Because the location split cannot properly test novelty, we ran the species hold-out (leave-one-species-out) on the already-scored test species — a free preview that needs no new inference. With a *graded* novelty axis, the taxonomic effect **collapses to essentially zero** (Table 4), confirming that the location split's −0.36 was an artifact. The out-of-sample gain is again not significant — indeed slightly *negative* (the distance model is no better than, or marginally worse than, predicting the mean).

Factor	pDetA coef. [95% CI]	pAssA coef. [95% CI]
Taxonomic distance	+0.026 [−0.31, 0.31]	+0.008 [−0.33, 0.31]
Visual distance	+0.120 [−0.23, 0.38]	+0.135 [−0.23, 0.40]
Environment distance	−0.025 [−0.30, 0.21]	−0.046 [−0.33, 0.19]
Clutter	−0.106 [−0.37, 0.21]	−0.135 [−0.41, 0.20]
Night / IR	−0.394 [−0.99, 0.09]	−0.466 [−1.09, 0.06]
log(support)	−0.188 [−0.49, 0.19]	−0.193 [−0.51, 0.21]

Table 4: Split A (species hold-out, test-species pool) standardised coefficients. With a graded novelty axis the taxonomic and visual effects are ≈ 0 and span zero. pseudo-R^2 is 0.033 / 0.039.

Split	Hold-out	Target	ΔMAE	95% CI	p
Location (B)	location	pAssA	+0.007	[−0.005, 0.018]	0.13
Location (B)	location	pDetA	+0.004	[−0.007, 0.015]	0.23
Species (A)	species	pAssA	−0.003	[−0.013, 0.006]	0.74
Species (A)	species	pDetA	−0.004	[−0.013, 0.005]	0.79

Table 5: Out-of-sample gain over a mean predictor (ΔMAE > 0 means the model is better), with a paired group-bootstrap CI and one-sided p. **No row is significant** on either split.

4.4 What the results mean (honestly)

On both splits tested, the label-free distances do **not** predict SAM 3's per-cell detection or association to a degree any valid, group-clustered test supports. This is **H0-leaning**. It is not yet "H0 confirmed for the whole thesis": the species hold-out preview uses only the 83 test species, and the powered version (adding a sample of train species) is running now. The two *large-looking* effects that survive as point estimates — night/IR and (on Split B) taxonomic — do not survive an honest interval, and the taxonomic one is a known artifact. The one robust qualitative signal is that **night/IR footage lowers both** pDetA and pAssA (day → night: pDetA 0.60 → 0.51, pAssA 0.62 → 0.52), but even this is not individually significant once clustering is respected.

5 Issues encountered and how we solved them

This section is the engineering and statistics log. Each row is a real problem we hit and the fix applied.

5.1 Engineering / infrastructure

5.2 Statistical / scientific

6 Current status and next steps

- **Done and validated:** the full measurement + modelling pipeline; the location split (Split B, null); the species hold-out preview (Split A on test species, cleaner null); honest clustered inference throughout.

Issue	Solution applied
Feature assembly ran out of memory at full sampling caps (all cropped frames held in RAM at once on a ∼58 GB container).	Rewrote the embedding step to work in *chunks*: load → embed → cache → free, so peak RAM is bounded by one chunk regardless of the caps. RSS stayed at ∼7 GB.
Embedding was I/O-bound at ∼5 crops/s (the GPU sat idle while single-threaded frame reads from a network filesystem blocked).	Parallelised frame load/crop with a thread pool (∼32 workers); the GPU is now the bottleneck.
The official scorer failed on a fresh pod (its import-time dependencies were missing).	Hardened the pod setup to clone the scorer and install its undeclared deps (`iopath, einops, timm, ftfy, regex`).
PyTorch pulled a CUDA build newer than the pod's driver; and `pip` was OOM-killed installing torch (the temp dir was RAM-backed).	Pin torch to the driver's CUDA; move `TMPDIR` onto persistent disk and install with `--no-cache-dir`.
The full train-split inference is ∼31,000 probes ≈ **10 days** of GPU time — far too expensive for one experiment.	Added a *per-species cap* (a stratified sample of ∼30 positive videos per species, ≈1500 probes ≈6 h) that still gives a graded novelty axis.
A single pathological clip crashed SAM 3's forward pass (a likely CUDA out-of-memory on a long video) and killed the entire multi-hour batch.	Wrapped per-video inference in a guard: a crashing clip is *skipped* (scored as a miss), the GPU cache is freed, and the batch continues.
Long clips could exhaust GPU memory during video preprocessing.	Keep SAM 3 video preprocessing on the CPU (raw storage only), running the model itself on the GPU.
The proxy SSH to the pod allows no `scp` and needs a pseudo-terminal.	Transfer files by `tar`+`base64` over the terminal and decode locally; run the analysis parquets locally where possible.

- **Running now:** the *powered* species hold-out — a capped (∼30 videos/species) SAM 3 pass over the train split (∼1500 probes, ∼6 h), which will add species and cells to tighten the H1 test.
- **Then, in priority order:** (i) refit Split A with the pooled train+test species; (ii) a multi-object-only association analysis; (iii) the hard-negative false-positive analysis; (iv) the familiarity proxy; (v) if the powered test confirms the null, pivot the thesis toward *representational probing* (does SAM 3 encode taxonomy?), which H0 explicitly motivates.

Bottom line. The engineering works end-to-end and the statistics are now honest. The current, carefully-tested answer is that simple label-free distances do not predict this tracker's zero-shot transfer on the splits measured so far — a clean, publishable negative result that sharpens the next question rather than a failure.

Issue	Solution applied (or proposed)
Over-precise confidence intervals. Weighting each cell by its frame count inflated the effective sample $\sim$85× and shrank every standard error $\sim$9×; on top of that, predictors are constant within a species/location (pseudo-replication), so 346 cells are really only $\sim$60–92 independent groups. The naive intervals looked "significant" when they were not.	Replaced the naive intervals with a **group-cluster-bootstrap**: resample whole species / locations, refit, take percentile intervals. Every distance CI then spans zero. *Next:* also report cluster-robust (sandwich) SEs as a cross-check.
No significance test on the cross-validation gain. A +0.005 MAE edge over the baseline was being read as "beats baseline" with no test.	Added a **paired group-bootstrap** on the CV gain (resample whole held-out groups) → a CI and one-sided p. All gains are non-significant.
Degenerate temporal feature. On the location split the temporal gap is present for only 32/346 cells and constant among them — an all-zero column after standardising.	Drop zero-variance predictors automatically. Temporal distance is therefore *untested* (not falsified) on this split; it needs a time-based split to test.
Novelty axis was degenerate on Split B. With the train split as reference, the taxonomic distance is ≈ 0 for shared species and > 0 for only $\sim$7 truly-unseen ones, so the "-0.36" coefficient was a fragile 7-species spike-at-zero, not a graded effect.	Run **Split A** (leave-one-species-out), where novelty is graded. The taxonomic effect then collapses to ≈ 0 — confirming the artifact.
Environment sign-flip. The continuous scene distance has a weak *negative* raw correlation but flips to $\approx$ 0/positive in the fit — a suppression symptom of its collinearity with the night/IR flag (both are per-location, from the same background).	Flagged; VIF currently covers only the four distances. *Next:* add night/IR and clutter to the VIF check and refit environment with/without night/IR to show the confound explicitly.
Detection-vs-association is under-identified. 78% of cells are single-object, so pAssA$\approx$ pDetA and the two regressions are near-duplicates — the project's core decomposition cannot be tested here.	*Next:* a targeted analysis on the $\sim$22% multi-object cells, where association carries information independent of detection (flagged exploratory, low power).
Hard negatives excluded from the fit. The detection model is fit on animal-present cells only (recall-given-present), so the	*Next:* a separate false-positive-rate-vs-novelty analysis that reintroduces the 1101 hard negatives.